MIL-A-47089(MI)
1 May 1974
SUPERSEDING
MIS-12799A
5 October 1970
MPD-9148B
5 September 1961

MILITARY SPECIFICATION

ADHESIVE, METAL FILLED, CONDUCTIVE, ELECTRICAL AND THERMAL

This specification is approved for use by all activities of the Department of the Army.

1. SCOPE

1.1 This specification covers one type of metal-filled adhesive that is electrically conductive.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250/4	Aluminum Alloy, 2024, Plate and Sheet
PPP-T-45	Tape, Gummed Paper, Reinforced and Plain, For Sealing and Securing
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

FSC 8040

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PPP-B-636	Box, Shipping, Fiberboard
PPP-C-96	Cans, Metal, 28 Gage and Lighter
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-76	Tape, Pressure-Sensitive, Adhesive, Paper, Water Resistant

Military

MIL-S-8516	Sealing Compound, Polysulfide Rubber, Electric Connectors and Electric Systems, Chemically Cured

STANDARDS

Federal

FED-STD-175	Adhesive, Methods of Testing
FED-STD-406	Plastics, Methods of Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specifice procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606).

American Trucking Association Publication

National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington, DC 20013)

3. REQUIREMENTS

3.1 First article. When specified by the procuring activity (see 6.2), the adhesive furnished under this specification shall have been inspected and passed the first article inspection specified herein (see 4.6).

3.2 Materials. The adhesive covered by this specification shall consist of a silver-filled, epoxy-type resin and a separate type hardener. Cure shall be effected by catalytic action and shall not depend upon evaporation.

3.2.1 Shelf-life. The manufacturer of the adhesive shall certify that the adhesive will meet all the requirements of this specification after storage for 6 months in the original, unopened containers at a temperature not exceeding 23 degrees centrigrade (C), 73 degrees Fahrenheit (F).

3.3 Physical properties of uncured adhesive. The adhesive, when mixed in accordance with the manufacturer's instructions, shall meet the physical property requirements specified herein.

3.3.1 Pot life. The adhesive shall have a minimum pot life of 15 minutes (for 20 gram batch) at 23 plus or minus 2 degrees C (73 plus or minus 4 degrees F).

3.4 Workmanship. The adhesive shall be homogeneous and free from foreign material. Containers shall be clean, uniformly filled, well sealed, and legibly marked.

3.5 Low-temperature storage. The cured adhesive shall meet the requirements of this specification after storage for 72 hours at minus 62 plus or minus 3 degrees C (minus 80 plus or minus 5 degrees F).

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3.6 Physical properties of cured adhesive. The adhesive, after curing 24 hours at 23 plus or minus 2 degrees C (73 plus or minus 4 degrees F), or curing 4 hours at 66 plus or minus 2.8 degrees C (150 plus or minus 5 degrees F), shall meet the physical property requirements of Table I.

Table I. Physical Properties of Cured Adhesive

Property to be tested	Conditioning Procedure (see 4.7.4.1.2)	Property Value	Test Method	
			Reference	Modified by Paragraph
Volume resistivity maximum, ohms per cubic centimeter	A E-1/165 E-1/-65	0.001 0.001 0.050	FED-STD-406 Method 4041	4.7.4.2.2
Shear strength, minimum, pounds per square inch	A E-1/165 E-1/-65	2,200 2,200 2,200	FED-STD-175 Method 1033.1	4.7.4.2.3

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Verification inspection. All quality assurance operations performed by the contractor will be subject to Government verification at scheduled or unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied and (b) Government product inspection to measure the quality of the product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect on the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with the operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.3 Definitions. The definitions of inspection terms shall be in accordance with MIL-STD-109.

4.4 Inspection equipment. The Government inspector, when examining the contractor's inspection equipment, will determine that the contractor has available, and utilizes correctly, test equipment of sufficient accuracy, quality, and quantity to permit performance of the required inspections. The contractor shall have available a set of master gages, standards, and appropriate instruments to conduct regularly scheduled calibrations of his inspection equipment. Records of such calibrations shall be maintained and made available for Government review. The calibration of gages, standards, and instruments will be checked periodically by authorized Government personnel.

4.5 Classification of inspection. The classification of inspection shall be as follows:

a. First article inspection

b. Acceptance inspection

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4.6 First article inspection.

4.6.1 First article sample. The first article sample to be inspected under this document shall be representative of the subsequent production and shall meet all the requirements as specified herein.

4.6.2 Sampling. Sampling for first article inspection shall be specified by the procuring activity.

4.6.3 Inspection site. Unless otherwise specified, the first article inspection site shall be as outlined in 4.1.

4.6.4 Examinations. The first article examinations shall consist of all the acceptance examinations specified herein.

4.6.5 Testing. The first article tests shall consist of all the acceptance tests specified herein plus the following test.

4.6.5.1 Low-temperature storage. The cured adhesive shall be placed in a temperature chamber at a temperature of minus 62 plus or minus 3 degrees C (minus 80 plus or minus 5 degrees F) for a minimum of 72 hours. Upon completion of the 72-hour exposure, the adhesive shall be allowed to stabilize at normal room temperature and then shall be tested for conformance with all the acceptance tests specified herein.

4.6.6 Rejection and resubmittal. Unless otherwise specified, samples which do not meet all requirements of the first article inspection will be rejected and returned to the contractor. Before resubmittal, full particulars concerning previous rejection and the action taken to correct the defects shall be furnished the procuring activity.

4.6.7 Instruction sheet. Along with the first article sample, the manufacturer shall forward an instruction sheet detailing complete mixing instructions.

4.7 Acceptance inspection.

4.7.1 Lot size. Lot size shall not exceed that quantity of adhesive produced in one continuous operation by the same manufacturer and manufactured in accordance with the same specification, specification revision, and other requirements. Each lot shall be composed of raw materials from homogeneous lots.

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4.7.2 Sampling.

4.7.2.1 For examination. Sampling for examination shall be 100 percent of production except as specified in 4.7.2.1.1.

4.7.2.1.1 Sampling for inspection of containers. A random sample of filled containers shall be selected by the Government inspector in accordance with MIL-STD-105 at inspection level I and acceptance quality level at 2.5 percent defective to verify compliance with all requirements regarding fill, closure, marking and other requirements not requiring tests.

4.7.2.2 For testing. Unless otherwise specified, testing shall be performed on a sample taken at random for each lot. The sample shall be of a size sufficient for the performance of the required tests and shall be marked with the manufacturer's name, product number, lot number, and date of manufacture.

4.7.3 Examinations. Visual examination of the sample specified in 4.7.2.1 shall be conducted for the purpose of determining compliance with the requirements for construction and workmanship, as well as the requirements for preservation, packaging, packing and marking covered in section 5.

4.7.3.1 Inspection of containers. Each sample filled container shall be examined for defects of construction of the container and the closure for evidence of leakage and for unsatisfactory marking; each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for Government inspection provided the contractor has removed or repaired all nonconforming containers.

4.7.4 Testing.

4.7.4.1 Test methods and procedures.

4.7.4.1.1 Test specimens. Five test specimens shall be prepared in accordance with the respective test method required for each test specified in table I. Test specimens requiring cure shall be cured before testing, in accordance with 3.6 for the applicable cure cycle.

4.7.4.1.2 Conditioning. Prior to testing, specimens shall be conditioned as specified in table I.

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4.7.4.1.2.1 Designation. The type of conditioning required shall be designated as follows:

Condition A: As received; no special conditioning.

Condition E: Temperature conditioning.

4.7.4.1.2.2 Procedures. The conditioning procedure for condition E shall be indicated by the following combination symbols:

a. A capital letter indicating the type of conditioning.

b. A number indicating the duration of the conditioning in hours.

c. A number indicating the conditioning temperature in degrees F.

The numbers shall be separated from each other by a slant mark and from the capital letter by a dash.

4.7.4.1.2.3 Conditioning tolerance. Unless otherwise specified, conditioning tolerances and additional testing information shall be as specified in Table II.

4.7.4.2 Tests. Tests shall be as specified in table I with the following stipulations:

4.7.4.2.1 Pot life. Pot life shall be tested in accordance with the requirements of the paragraph entitle "Application Life" of MIL-S-8516.

4.7.4.2.2 Volume resistivity. Volume resistivity shall be tested in accordance with method 4041 of FED-STD-406 at 23 plus or minus 2 degrees C (73 plus or minus 4 degrees F) and 50 plus or minus 5 percent relative humidity. Volume resistivity shall be determined on test specimens cured in accordance with both the cure cycles specified in 3.6.

4.7.4.2.3 Shear strength. Shear strength shall be tested in accordance with method 1033.1 of FED-STD-175, using aluminum adherends conforming to QQ-A-250/4 and having a T3 temper. Shear strength shall be determined on test specimens cured in accordance with both the cure cycles specified in 3.6.

4.7.5 Rejection and resubmittal.

4.7.5.1 Any sample which does not meet all the acceptance inspection requirements shall be rejected. Rejected samples shall be properly identified and stored under such conditions as to preclude any possibility of the loss of identification. Rejected samples may be reworked when an applicable approved blending or reworking procedure exists, or when the reworked product will not deviate from the specification or other applicable documents.

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Table II. Conditioning Tolerances and Testing Information

Conditioning Procedure	Conditioning Tolerances		Testing Conditions	
	Time	Temperature	Temperature and Tolerance	Remarks
E (low temperature)	-0, +6 minutes	±5 degrees F	-55 ±5 Degrees F	Start test immediately after the cycling has been completed.
E (high temperature)	-0, +6 minutes	±5 degrees F	+165 ±10 Degrees F	Start test immediately after the cycling has been completed.

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4.7.5.2 Rejected samples which are not reworked in accordance with 4.7.5.1 above shall be processed in accordance with specific instructions from the procuring activity.

4.7.5.3 Before final acceptance, full particulars concerning previous rejection, the action taken to correct the defects found in the original sample, and evidence of corrective action in current production shall be available to the procuring activity.

5. PREPARATION OF DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Unit packaging. The metal-filled adhesive shall be enclosed in metal containers in accordance with PPP-C-96.

5.1.1.2 Intermediate packaging. When intermediate packaging is required, unit packages shall be placed in close-fitting containers conforming to Class WR of PPP-B-636. Separaters, partitions, and filler strips conforming to PPP-F-320 shall be used. The intermediate containers shall be closed as specified in 5.1.1.3.

5.1.1.3 Closure of fiberboard boxes. Unless otherwise specified, containers conforming to PPP-B-636 shall be closed with tape conforming to PPP-T-76.

5.1.2 Level B same as Level A.

5.1.3 Level C. Unless otherwise specified, preservation and packaging shall afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A. The metal-filled adhesive shall be packed in overseas-type exterior containers conforming to one of the following specifications: PPP-B-585; PPP-B-591; PPP-B-601; or PPP-S-621. The container shall be strapped in accordance with the applicable container specification.

5.2.2 Level B. The metal-filled adhesive shall be packed in domestic-type exterior containers conforming to one of the following specifications: PPP-B-585; PPP-B-591; PPP-B-601; or PPP-B-621.

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5.2.3 Level C. The metal-filled adhesive shall be packed for shipment in conformance with Uniform Freight Classification and National Motor Freight Classification.

5.3 Markings. Interior packages and exterior containers shall be marked in accordance with MIL-STD-129 as implemented by the using service. Necessary safety and toxicity precautions shall be marked on the interior containers.

6. NOTES

6.1 Intended use. The adhesive covered in this specification is intended to be used in making electrical connections when hot soldering is impractical, and in filling voids where electrical conductivity must be maintained.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether or not a first article sample is required (see 3.1).

c. Applicable sampling plan and level of inspection.

d. Selection of applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-12799, dated 5 October 1970 and Missile Purchase Description MPD-9148B, dated 5 September 1961.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Project No. 8040-A046

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

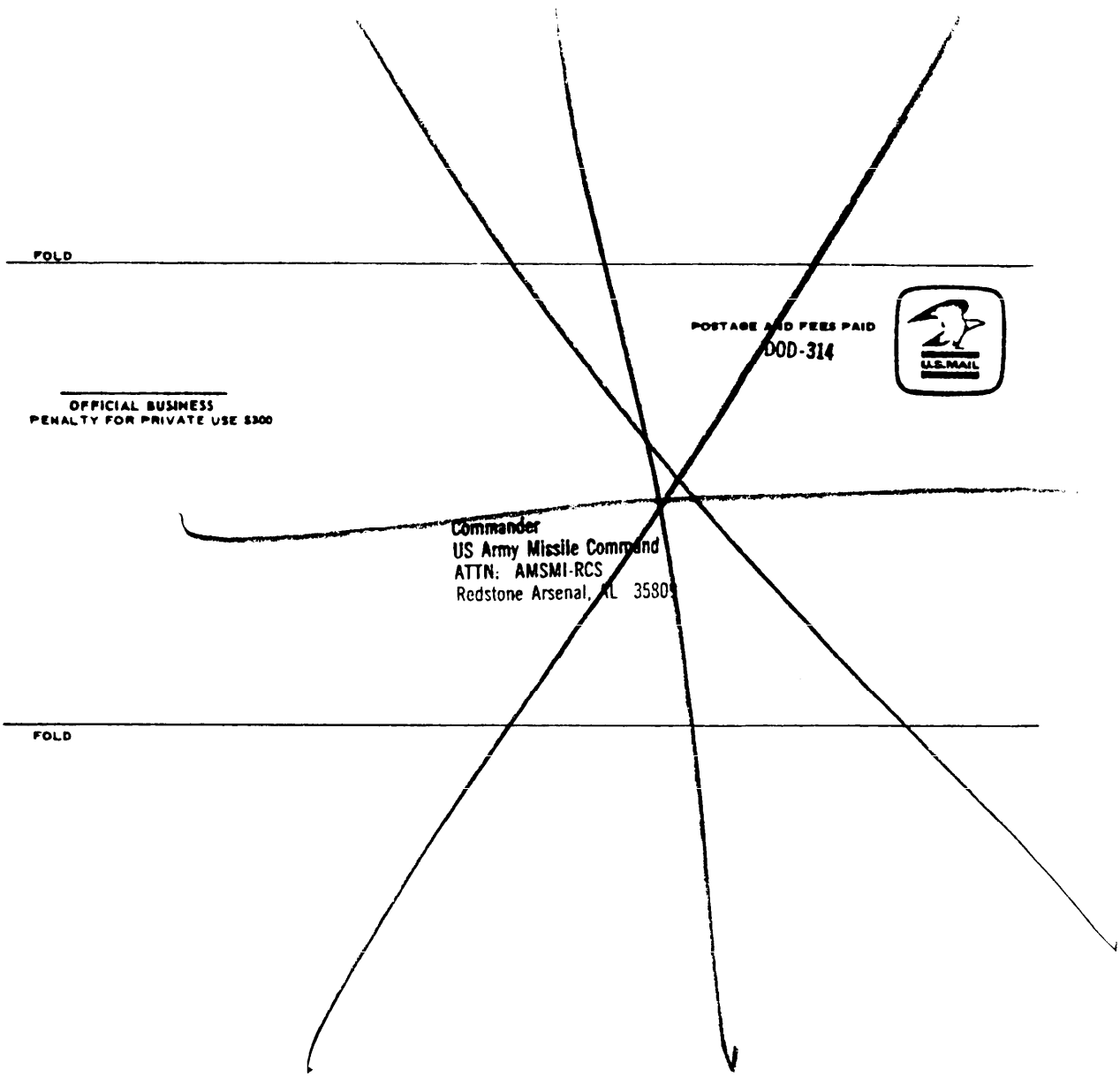
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